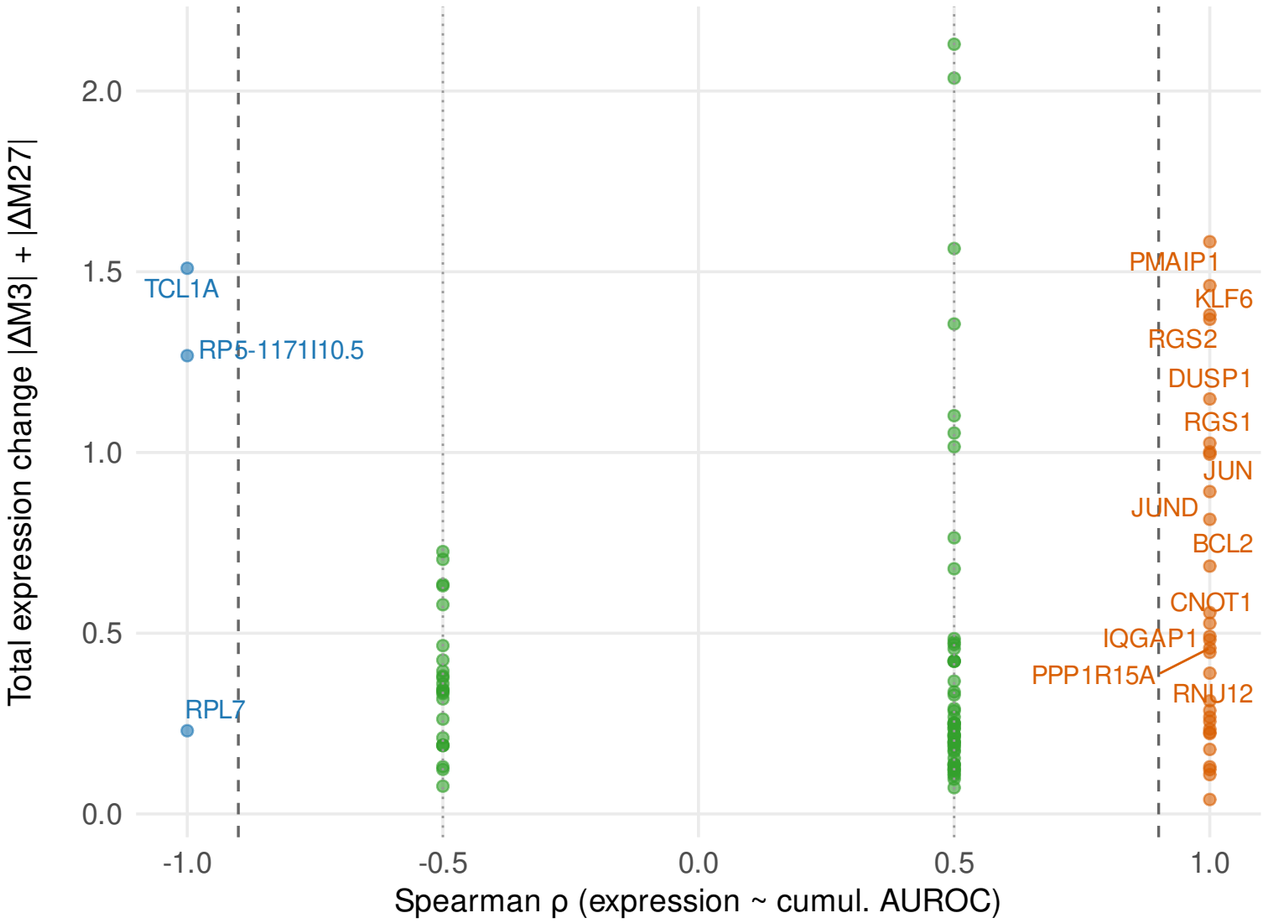
PK-expression correlation analysis: ibrutinib resistance in CLL

Top: Spearman ρ between expression/TF trajectories and cumulative AUROC. Bottom: AutoML R^2 performance.

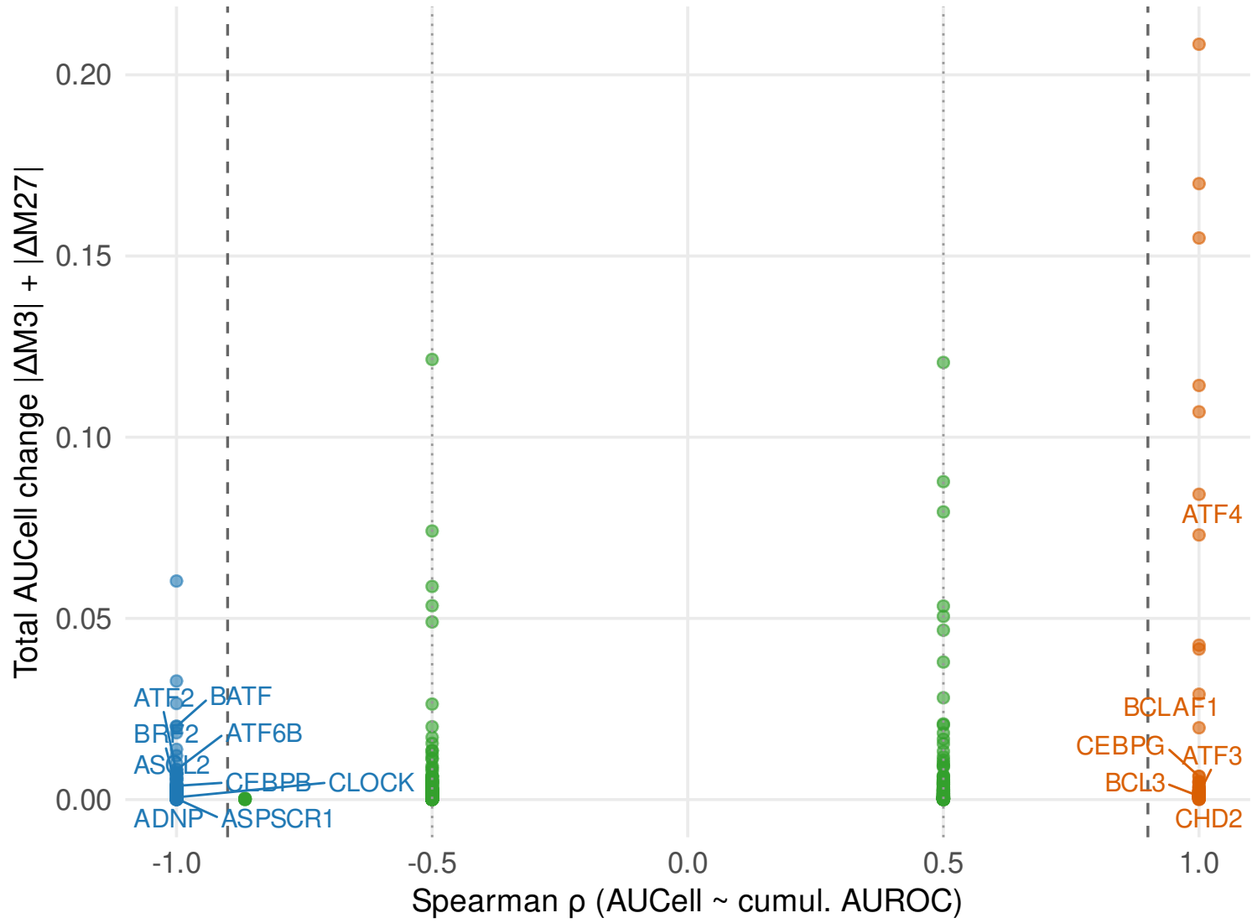
Gene expression trajectory vs cumulative AUROC

Spearman ρ between expression (M0→M3→M27) and cumulative ibrutinib AUROC



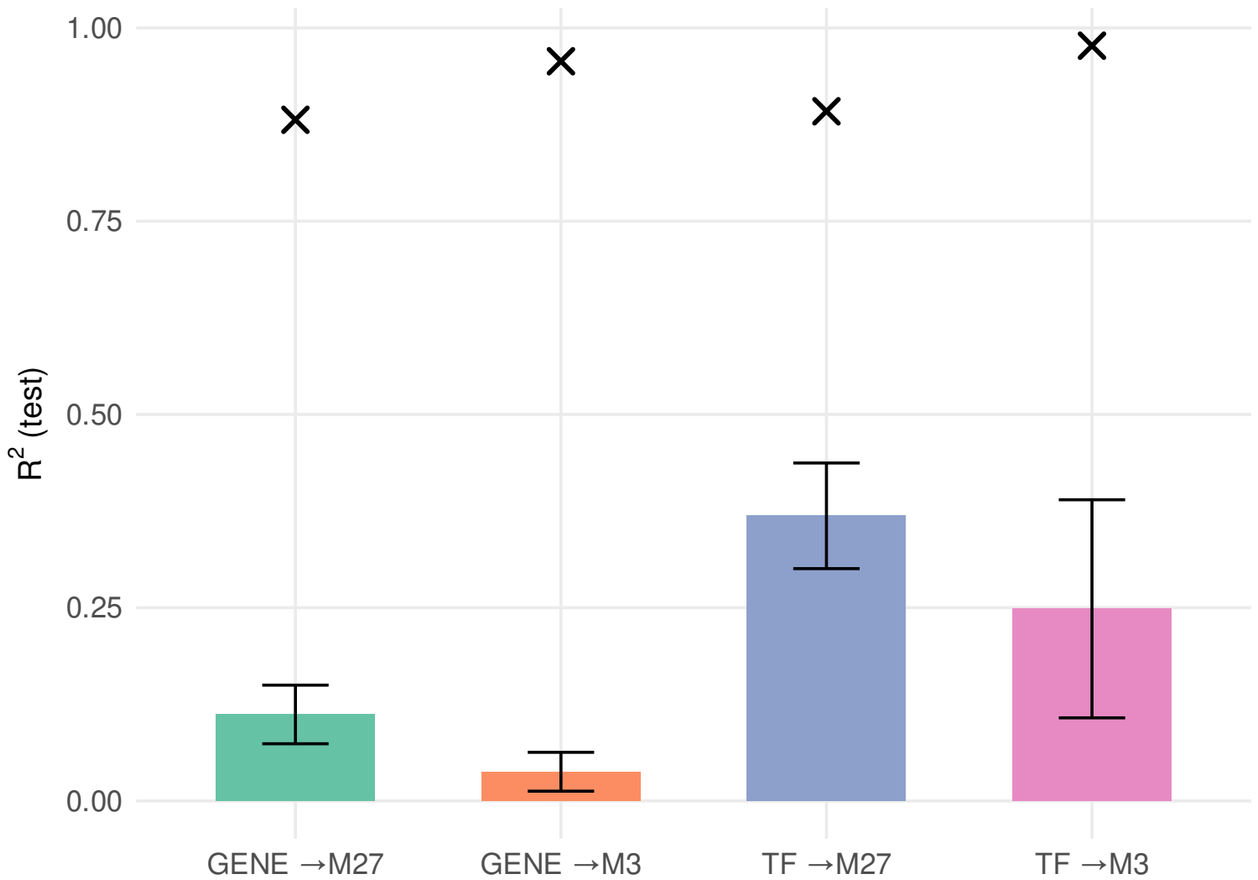
TF activity trajectory vs cumulative AUROC

Spearman ρ between AUCell score (M0→M3→M27) and cumulative ibrutinib AUROC



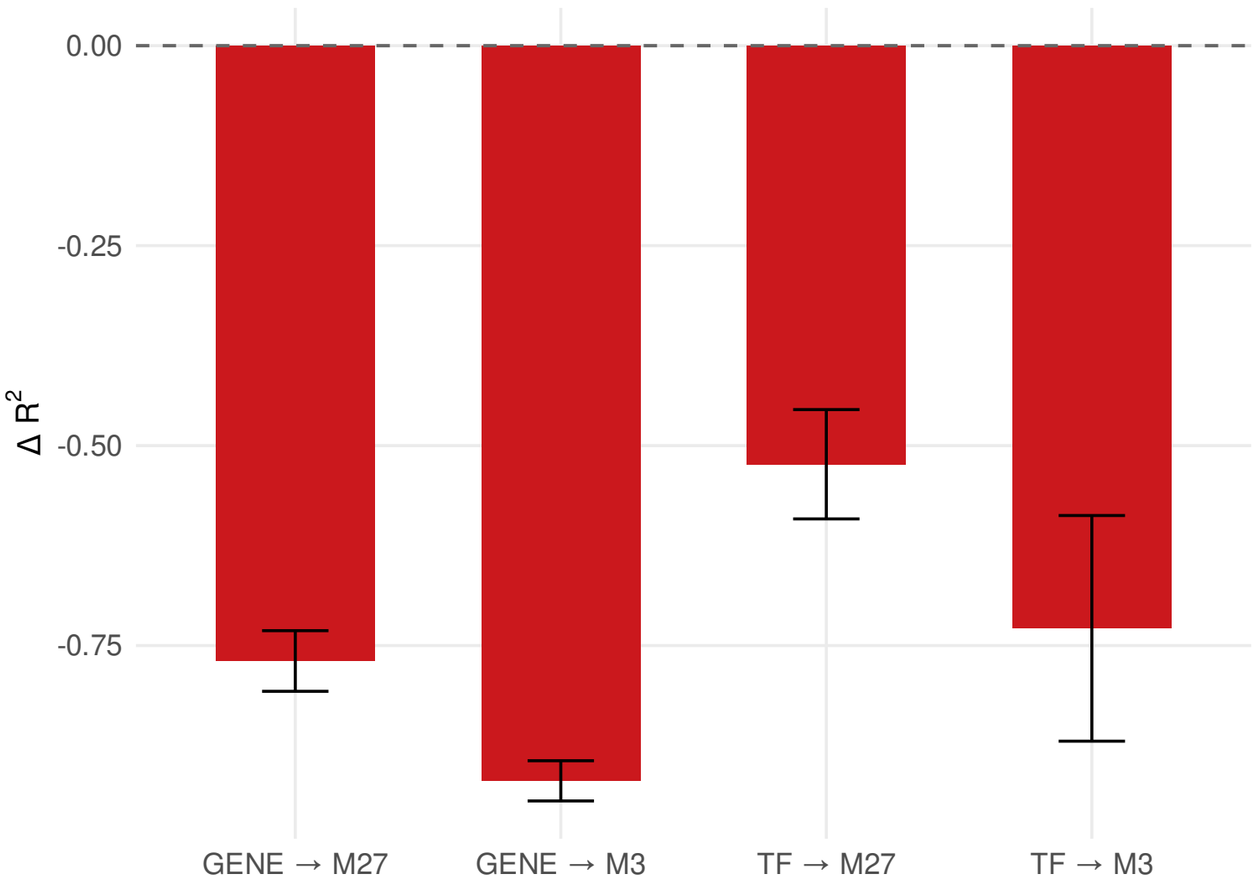
AutoML test R^2 vs M0 baseline

× = baseline-only R^2 | error bars = SD across noise replicates



PK contribution: ΔR^2 (AutoML – baseline)

Positive = drug exposure adds predictive value beyond M0



PK class Partial PK correlation | PK-driven (activated) | PK-driven (suppressed)

label GENE →M27 | GENE →M3 | TF →M27 | TF →M3